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A Holistic Overview of the Technical Challenges of CO₂ Sequestration in Depleted Hydrocarbon Reservoirs

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Abstract

While depleted hydrocarbon reservoirs (DHRs) may appear to be optimal for the purpose of geologic carbon (CO₂) storage (GCS), their low abandonment pressure poses a substantial challenge. Injecting high-pressure, low-temperature CO₂ into a low-pressure reservoir can induce various physical phenomena, particularly in the near wellbore region, including hydrate formation and thermal fracturing. These effects may be exacerbated by the Joule-Thomson (J-T) effect and are of critical importance for ensuring the sustainability of CO₂ injectivity and maintaining the integrity of both wells and caprock.

The near wellbore as well as integrity issues are crucial in using DHRs for GCS. Near the wellbore, the thermodynamic conflict between the warm low-pressure reservoir and the cold high-pressure CO₂ can lead to J-T cooling, potentially forcing the system into the hydrate stability zone. On the other hand, salt precipitation near the wellbore can also act negatively on the CO₂ injectivity. Cyclic CO₂ injection is likely to exacerbate the situations. The fracturing of the formation as well as well components induced by both thermal and pressure effects can cause a serious threat to the containment of CO₂ in the reservoir, especially in depleted gas reservoirs (DGRs).

This paper aims to discuss the technical challenges of GCS in DHRs based on the laboratory and numerical works conducted in our Institute. We discuss the formation and equilibrium conditions of the metastable zone of CO₂ hydrates formed in salt solutions in a specially designed setup. Dynamic experiments (coreflood) and numerical work on salt precipitation are also elaborated to identify the conditions that may aggravate the near wellbore issues. We report the results of the experiments carried out in a large-scale setup which mimics real wellbore environment in terms of cyclic pressure and temperature conditions.

The overall discussion provides a holistic technical view and categorization of the challenges and their potential mitigations that may be useful for the certification and operation of GCS projects in DHRs.

Keywords: Geologic carbon storage (GCS), depleted gas reservoir (DGR), wellbore integrity, cyclic CO₂ injection, CO₂ hydrates, salt precipitation, Joule-Thomson effect

Introduction

Globally, DHRs represent huge potential storage volumes for CO₂ (in the range of 675-900 Gt) offering unique geological characteristics that make them attractive for geologic carbon storage (GCS), including proven containment systems, existing infrastructure, and significant storage capacity (Wei et al. 2023). They present several compelling advantages for CO₂ storage (Alkan et al. 2022). The primary advantage is evident in the **demonstrated capacity** of these reservoirs to trap hydrocarbons over extended geological periods. This provides substantial evidence supporting the hypothesis that CO₂ will be securely contained (IEAGHG 2022; Wei et al. 2023). The caprock sealing performance has already been tested by nature, reducing uncertainty about long-term storage security. Secondly, **existing infrastructure** where much of the necessary infrastructure (wells, pipelines, surface facilities) can be repurposed for CO₂ storage operations, potentially reducing project costs (IEAGHG 2022). Thirdly, **geological characterization**; the extensive data from hydrocarbon production provides detailed knowledge of reservoir characteristics (porosity, permeability, structure) that is typically unavailable for saline aquifers (SAs) (IEAGHG 2022). Fourthly, **economic benefits** where CO₂ injection can enable EOR or enhanced gas recovery (EGR), creating revenue streams that may offset storage costs (Heidarabad and Shin 2024). Studies indicate CO₂-EOR can increase oil recovery by up to 40%. Fifthly, **reduced seismic risk**; some evidence suggests that depleted reservoirs may be less prone to injection-induced seismicity because pore pressure must first be restored to original levels before exceeding fracture thresholds (Samuels 2021).

There are, however, some important technical challenges related to injection into them. Upon high-pressure dense-phase CO₂ injection, the low pressure in the depleted reservoirs results in strong cooling and potential freezing of the wellbore and near-wellbore zone (NWZ), due to the Joule-Thomson (J-T) effect and associated phase changes of the CO₂. This risk is especially pertinent in the context of DGRs, where the depletion pressure can be significantly lower in comparison to that observed in oil reservoirs. This can jeopardize injectivity due to e.g., hydrate formation, as well as near-wellbore stability and well integrity.

This paper aims to discuss the technical challenges of GCS in DHRs based on the laboratory and numerical works conducted at our Institute as well as referring to recent literature on the topic. We report the results of the experiments carried out in a large-scale setup which mimics real wellbore environment in terms of cyclic pressure and temperature conditions. We discuss the formation and equilibrium conditions of the metastable zone of CO₂ hydrates formed in salt solutions in a specially designed setup. Dynamic experiments (coreflood) and numerical work on salt precipitation are also elaborated to identify the conditions that may aggravate the problem. The primary focus of this study is the DGRs, which comprise the majority of the DHRs available for GCS. A distinguishing feature of the DGRs is their standalone GCS opportunities, which differ from the depleted oil reservoirs in which GCS is applied as part of the continuum of CO₂-EOR operations.

Characteristics of GCS in DGRs; Storage Mechanisms and Applications

Similar to GCS in SAs, CO₂ is trapped in DHRs through multiple mechanisms that operate at different timescales. The main mechanism is **structural/stratigraphic trapping** where CO₂ is physically contained beneath impermeable caprock. This mechanism is generally supported by the residual/capillary trapping that occurs as CO₂ becomes immobilized as disconnected droplets in pore spaces due to capillary forces where brine saturation exists. **Dissolution trapping** occurs when CO₂ dissolves into formation water, however due to lower brine saturation this mechanism is of lower relevance in DGRs. **Mineral trapping** is a mechanism over long timescales, in which dissolved CO₂ react with reservoir minerals to form stable carbonate minerals, however with lower impact to total storage capacity of DGRs again due to low initial brine saturation. In DGRs, studies show that the majority of injected CO₂ remains mobile, with capillary and dissolution trapping playing minor roles.

In addition to international expertise development projects a number of CO₂ injection and demonstration storage projects using a DGR have been designed and (partly) conducted in the past, at a smaller (e.g., K12-B, Netherlands, or Lacq, France) or larger scale (e.g., Goldeneye, UK, and ROAD and Porthos, the Netherlands). These projects have provided valuable expertise and experience on the design of injection activities. The current industry-initiated research targets what operators see both as one of the main advantages, in terms of increased storage capacity, and the main challenge related to making depleted reservoirs widely available for large-scale storage, namely their low pressure. Whatever the pressure at the wellhead is, the liquid CO₂ column in well exert a significant pressure to the bottom hole and entrance of high-pressure CO₂ into a low pressure reservoir may result in strong cooling due to the J-T effect and potential freezing of the well and the NWZ. These effects could jeopardize injectivity, as well as near well formation stability and well integrity, in addition to the pressure impact on possible hydraulic fracturing. Potential solution is the heating of the CO₂ near the wellhead or downhole for enabling transport in the gas phase, instead of dense liquid phase, a solution which requires a large number of injectors to achieve sufficient injection volume with time. The solution is prohibitively expensive, particularly in offshore environments. In addition, this impacts the CO₂ footprint negatively – in contrast with its actual purpose. Therefore, novel solutions are required, namely the design of an injection well, or injection scheme, such that temperature and phase change effects are reduced to tolerable operational windows. Briefly stated, this means that it is necessary to understand how to ensure safe ‘cold CO₂ injection’ into depleted reservoirs.

Challenges of GCS in DGRs

In addition to the containment/integrity risks created from using old legacy wells use of DGRs for GCS presents some other challenges that may pose uncertainties and risks on the **capacity, injectivity and containment (CIC)** of the operations. As illustrated in Fig. 1, the following challenges still exist and need to be addressed before large-scale deployment of GCS in DGRs can occur:

- Cooling of the reservoir and J-T effect created by the colder and high pressure CO₂ injection;
- Hydrate formation due to the J-T effect that may reduce injectivity;
- Salt precipitation that may reduce the injectivity standalone or together with hydrate formation;
- Chemical reactions that may cause additional precipitations or dissolution;
- Thermal fracturing due to the strong changes in temperature;
- Hydraulic fracturing due to the combined effect of temperature and pressure.

Because of the abrupt changes in thermodynamics as explained previously, these challenges occur mostly around the well, therefore the NWZ is especially important for the assessment and mitigation.

In the last few years, intensive research activities have been conducted to enable CO₂ storage in DGRs. Our institute was not an exception and we have been involved in two major international projects that were dedicated for CO₂ storage in DGRs. The first project INJECTWELL ran between November, 2021-December, 2023 (Burachok et al. 2024). The second project is RETURN Project (Amro et al. 2024). In both projects the above mentioned challenges were tackled at small and large scale. In the following sections, the main findings in the areas of cooling, hydrate formation, salt precipitations and wellbore integrity issues are reported and discussed in terms of their impact on CIC concerns as a holistic approach based on our physical and numerical modelling studies.

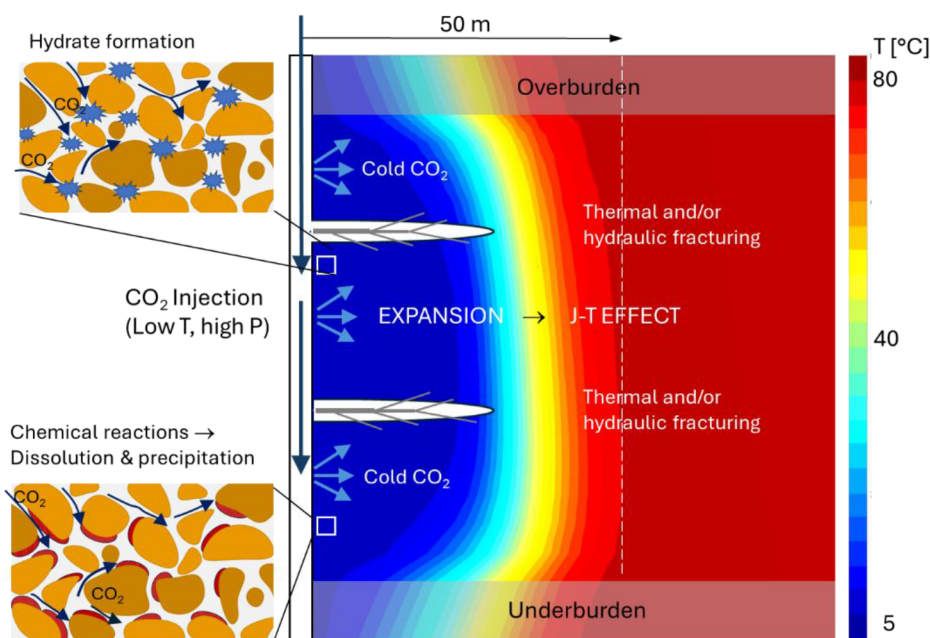


Figure 1—Near wellbore effects induced by low T and high P when CO₂ injected into a DGR with original reservoir temperature of 80 °C after 100 days of injection with 0.25 Mt/year; schematically described with a generic numerical simulation performed using TOUGH2.

Reservoir Cooling and J-T Effect

In general, the GCS operations may not be conducted isothermally. That is to say, the injection temperature may differ from the reservoir temperature significantly. Due to the volumetric concerns, cold CO₂ injection (higher density) is preferred. This results in various challenges in GCS. In the first one, the cold front progressing in the reservoir during the operation behind the CO₂ front cools down the reservoir. Fig. 1 depicts a screenshot of the temperature profile of a radial, 20 m thick GCS model in which CO₂ injection of temperature 5 °C is modelled with the GCS module of the numerical reservoir simulator TOUGH2. As can be seen, with an injection rate of ca. 8 kg/s (corresponding to 0.25 Mt/year which is at lower industrial range), the cooled front down to approximately 5 °C reaches 50 m away from the well just after only 100 days of injection. This change in the reservoir temperature is especially important in DGRs as the CO₂ transport is faster than the case of SA and can cause issues in all three pillars (CIC) of the GCS. Due to the strong dependency of CO₂ density to the changes in temperature, the capacity and injectivity calculations should consider the cooling of the reservoir. The sudden and strong changes in temperature can also cause integrity issues potentially due to the thermal fracturing.

Another additional cooling can arise due to the J-T effect in the NWZ. The injection of CO₂ results in a rapid quasi-adiabatic expansion leading to strong cooling (in some cases down to sub-zero temperatures). An example of this cooling effect upon CO₂ expansion can occur in the NWZ during the GCS in DGRs. In addition to theoretical considerations field reports support this challenge (e.g., Kubus 2009). The J-T effect in DGRs is investigated numerically as illustrated in Fig. 2 (Mathias et al. 2010; Pan and Oldenburg 2014; Hoteit et al. 2019; Gao et al. 2020). As can be concluded, especially for low reservoir pressures, temperatures, and permeabilities, low-temperature CO₂ injection may result in very low temperatures, even reaching hydrate stability zone (HSZ). This and similar numerical studies cannot exclude the formation of CO₂ hydrates under various thermodynamic and petrophysical conditions of GCS in DGRs therefore the topic and its potential mitigation has been one of the related topics.

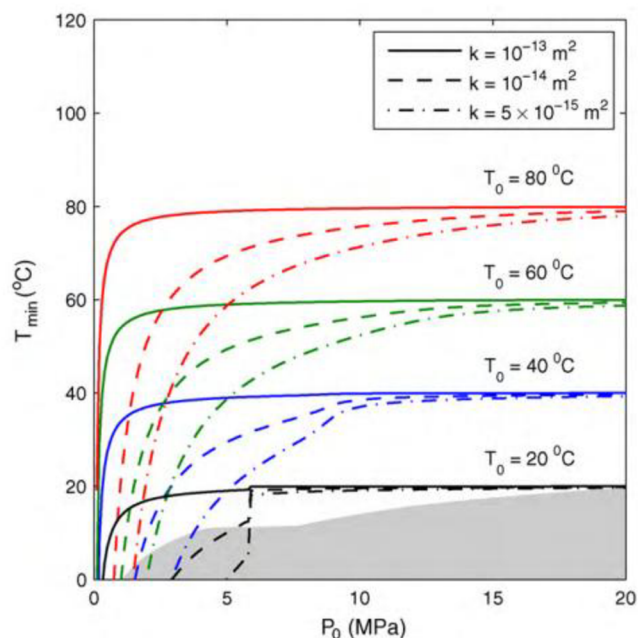


Figure 2—J-T effect: Hydrates can form in the NWZ (T_{inj} : injection temperature; T_0 and P_0 : reservoir temperature and pressure; k : reservoir permeability) (after Mathias et al. 2010).

Hydrate Formation

Hydrate formation due to the J-T cooling in the NWZ may be a potential challenge in the GCS in DGRs especially on the sustainability of the required injection rate as this can plug the pores reducing the injectivity.

We investigated experimentally the effect of temperature, pressure, salinity, gas impurities, time (downtime nearby hydrate formation conditions), porous media, and inhibitors on hydrate formation and dissolution using HYEX setup depicted in Fig. 3 and described in details in Tamáskovics et al. (2023, 2025).

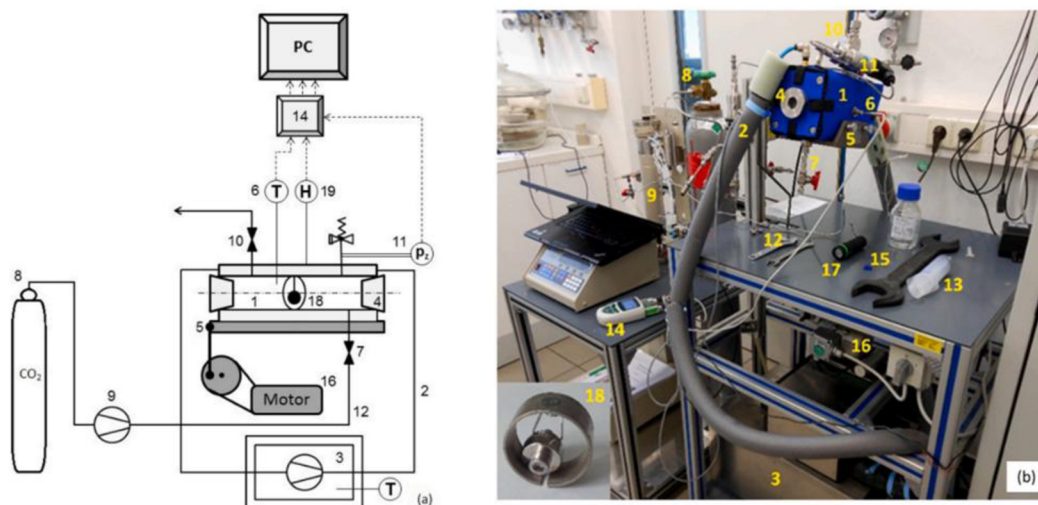


Figure 3—Hydrate experimental setup (HYEX) used to define hydrate stability zone in autoclave and porous media along with its inhibition. (a) schematic of the setup (b) picture (after Tamáskovics et al. 2023).

Fig. 4 compares the experimental results for CO₂ hydrate formation obtained using the HYEX setup. These results are plotted alongside the predicted curves from the HYDOFF model (Sloan 1998). The

difference in temperature between the equilibrium line and the hydrate formation points (represented by empty symbols) indicates the extent of the metastable zone of CO₂ hydrates.

The primary conclusions of our research on hydrate formation and its inhibition can be outlined as follows:

- Hydrate formation occurs at temperature higher than the freezing temperature of the liquid;
- As also observed in other studies (e.g., [Burgass et al. 2023](#)) salinity has an inhibitory effect on brine formation; the higher the formation salinity, the smaller the HSZ;
- The impurities in the CO₂ stream with N₂, synthetic air or CH₄ up to an amount 10 vol% have negligible effect on the HSZ;
- HSZ in porous media shifts to the left (colder temperatures) compared to batch experiments, while the dissociation shifts to the right. This is due to the heat transfer (cooling) into the matrix and slow diffusion of gas molecules in the porous medium; it takes longer time to form hydrate in porous media (kinetic shifting);
- In metastable region, hydrates are expected to form and therefore, inhibitors or other mitigations should be considered;
- Testing several thermodynamic inhibitors indicated that methanol is a suitable inhibitor for CO₂ hydrates; however, laboratory tests are essential under real reservoir conditions;
- After hydrate formation/dissociation in the porous sandstone cores, no fractures could be observed visually or via micro CT scan in the cores;

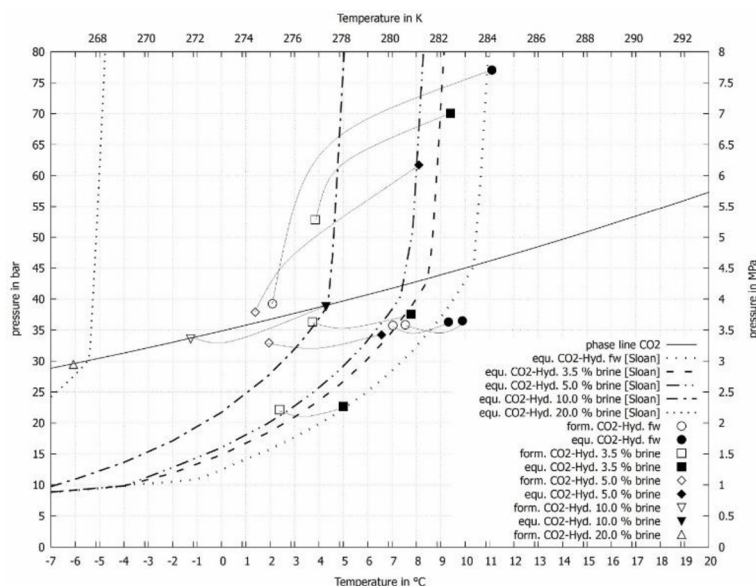


Figure 4—Pure CO₂ gas hydrate at different water salinities compared to Sloan (1998) reference data (after [Tamáskovics et al. 2023](#)).

Chemical Reactions – Salt Precipitation

The interaction between CO₂ and brine, specifically the dissolution of CO₂ in formation brine, is a pivotal parameter for all three pillars of GCS. The presence of CO₂ dissolved in brine has been shown to disrupt the geochemical equilibrium within the reservoir, thereby inducing the dissolution and/or precipitation of reservoir minerals. Carbonate mineralization is considered to be one of the mechanisms that ensure long-term trapping. However, alterations in mineral composition result in petrophysical changes, leading to increases or decreases in porosity and permeability, which directly impact storage integrity. The integrity of well components is susceptible to degradation due to dissolution or precipitation processes that occur

within the well and its interface with the surrounding formation. Nonetheless, in DGRs, it is anticipated that the brine saturation is minimal, provided that the depleted gas is not replaced by geologic water, such as water influx. In this case, the impact of geochemical and petrophysical alterations would likely be negligible. Within this context the NWZ of DGRs should be more emphasized because of the reasons discussed previously.

Salt precipitation in the NRW is one of the main concerns that may hinder the injectivity of CO₂. The threat in the use of SAs for GCS is obvious (e.g., Cui et al. 2023; Sun et al. 2025); the case in DGRs is less investigated. We investigated the case based on physical and numerical experimentation. Several flooding experiments were conducted in a special designed and constructed core flooding equipment. Complementary numerical studies were also conducted to compare to the core flooding results and deepen the knowledge. These were conducted using the reservoir simulator TOUGH2 and its corresponding EoS modules (Pruess et al. 1999; Pan et al. 2005).

A result of the experiment conducted at 60 °C while maintaining atmospheric pressure is given in Fig. 5. The NaCl based brine (17.5% wt.) saturated core was flooded with CO₂ at a constant flow rate of 3 ml/min. Gas breakthrough was detected after approximately 1 PV by decreased differential pressure, initiating the evaporation phase. The brine displacement phase occurred during the first 2000 PV flooded, after which brine displacement continued along with measurable evaporation due to decreasing differential pressure. At that time, the amount of water remaining in the core was about 36%, which is higher than what is usually reported for DGRs. Salt precipitation began after approximately 9 days of injection (4600 PV), as indicated by an increase in differential pressure. Differential pressure increased exponentially during the precipitation phase, while the permeability decreased toward blockage. Permeability decreases due to increasing salt precipitation, plugging of pores began after about 13.5 days of injection at about 7100 PV. An acceptable agreement with the experimental data runs was obtained in the numerical study of the case. Fig. 5 compares the calculated differential pressure and brine outflow with the experimental data. Although there is a significant difference in the values, the sudden increase in pressure is captured by the numerical simulation with about one-day difference (earlier). After this point, the differential pressure increases sharply, whereas in the experiments the increase is more steady and even gradual. Whether this is an experimental bias, a transition of the flow regime, or particle migration is not obvious. Salt precipitation was clearly visible at the inlet of the core after removing the sample out of the core holder. Since no precipitation was visible at the outlet side, it can be assumed that the precipitation is occurring/beginning near the inlet as also supported by numerical work. It should be noted that in all core flooding experiments, the CO₂ was injected at a temperature equal to experiment temperature i.e., injection temperature was equal to reservoir temperature. The numerical runs resulted in worse cases for each experiment with faster and higher salt precipitation than the experiments (Fig. 5). This may be due to the different parameters that need to be calibrated, such as the porosity-permeability relationship and the capillary pressure curves. The results have been confirmed through repeated experiments that used CO₂ in a liquid state (experiments at high pressure and high temperature) as well.

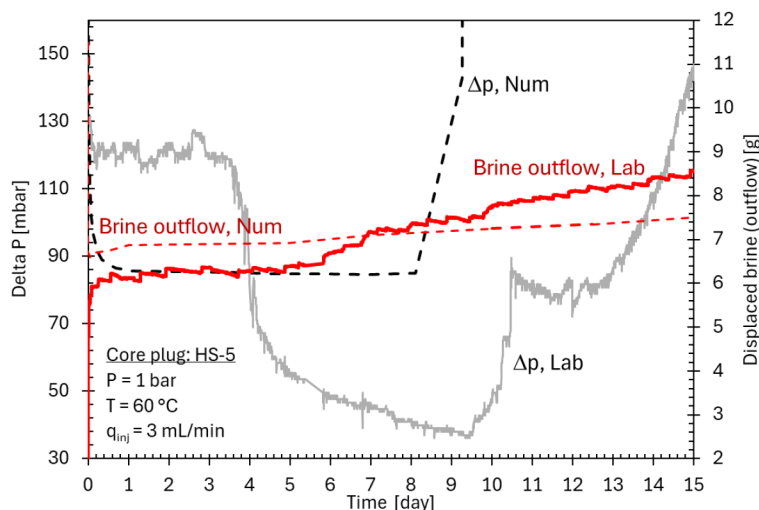


Figure 5—Comparison of observed and calculated differential pressure and brine production at the outlet for core HS-5, shown as an example from the salt precipitation experiments.

It was difficult to derive a relative permeability scheme from the experiments - not possible in part because of the challenges mentioned above. However, one result is clear and expected: the residual brine saturation at breakthrough is quite high for DGRs. In any case, this high saturation at the beginning of potential CO₂ injection cases results in higher injection times to observe the salting-out (SO) also increasing the probability of effective salt precipitation in terms of permeability impairment. On the other hand, continuous evaporation can reduce this saturation further with production/injection in DGRs increasing the injectivity, unless there is an additional brine supply to the reservoir, such as water influx. The accompanying numerical study focused on the missing aspects of the core flooding experiments, namely the use of more realistic parameters such as lower residual water saturation and colder CO₂ injection.

A generic DGR model with a thickness of 16 m was created with radial geometry where a fine grid discretization is applied around the well situated in the middle. The injection rate is selected equal to 0.5 Mt/year in all cases. The porosity and horizontal permeability of the reservoir are 0.2 and 100 mD successively. The initial brine saturation is assigned as 20% and the dissolved salt (NaCl) concentration is assumed to be 20% wt. Fig. 6 depicts the results as the permeability reduction profiles in NWZ (defined by the ratio of actual to absolute permeability) after two years of CO₂ injection for three different cases. As illustrated in Fig. 6, the cold CO₂ injection (10 °C) scenario (a) results in a substantially diminished extent and amount of salt precipitation compared to the isothermal case (b), wherein CO₂ is injected at reservoir temperature. This result is consistent with the established scientific understanding that colder CO₂ has a lower evaporation capacity compared to warmer CO₂. The permeability reduction is approximately 0.93 in both cases. The intersection with the underburden layer, where the initial brine saturation is 100%, is highlighted in the inset image. As demonstrated in this image, the permeability of the underburden blocks adjacent to the well exhibits a high level of reduction, with values less than 0.1 of the original permeability. Fig. 6 (c) highlights a case which can be called as an extreme case in terms of salt precipitation in DGR. The process entails the injection of warm CO₂ (at reservoir temperature) into a DGR with higher initial brine saturation, characterized by a lower porosity (0.05) and permeability (10 mD). In a period of two years, the permeability reduction in the reservoir was calculated to be 0.92, whereas the reduction in the reservoir layer adjacent to the underburden was approximately 0.5, exhibiting a significant decrease close to the well. The results, even in the extreme case, suggest no indication of injectivity restriction; instead, there appears to be an increase resulting from evaporation. Another positive aspect is the salt accumulation in the confining layers which may act as another barrier for CO₂ leakage. To investigate whether this picture may change with time, the model was run for 10 years. No significant change was observed except the extension of salt precipitation

in the reservoir as well as in the underburden blocks and adjacent reservoir blocks towards deep in the reservoir without any change in the injectivity.

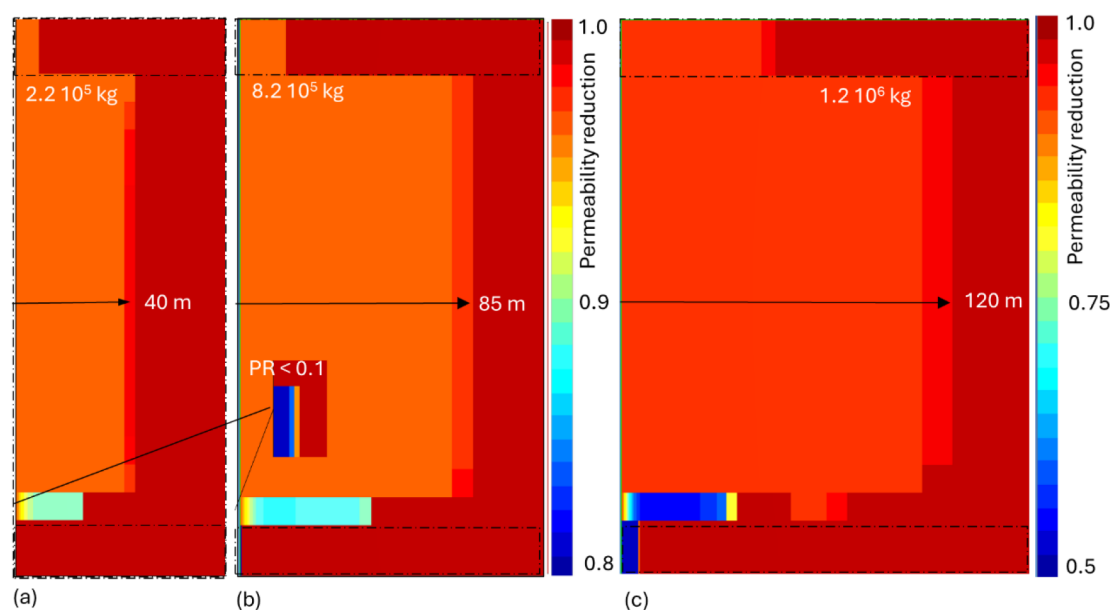


Figure 6—Permeability reduction profiles in NWZ (defined by the ratio of actual to absolute permeability) after two years of CO₂ injection for three different cases; (a) the injection temperature is 10 °C; (b) CO₂ is injected at reservoir temperature (80 °C); (c) the dissolved salt (NaCl) concentration increased to 25 % wt., reduced porosity and permeability for the injection temperature equal to reservoir temperature (extreme case).

However, based on the present study, it is felt that even when using DGR for GCS, salt precipitation can be an injectivity issue for the higher brine salinities and favorable geologic (e.g., water influx from the boundaries), petrophysical (e.g., lower porosity and permeability as well as high capillary pressures), thermodynamic (e.g., favorable pressure and temperature) and operational (e.g., low injection rates or cyclic injection). Also susceptible are cyclic injection cases where the periodic back and forth movement of the brine can be a reason for increasing salinity and thus precipitation. The use of numerical studies in the preliminary evaluation of potential salt precipitation is a necessity also considering the uncertainties of the experimental work. The calibration of the numerical model and the better interpretation of the experimental results would lead to a reduction of the uncertainties.

Thermal/Hydraulic Fracturing - Wellbore Integrity

The formation of thermal fractures is contingent upon the occurrence of extensive cooling of the reservoir, as has been discussed previously. The formation and/or opening of fractures may exert an influence on the NWZ region with a decline in bottomhole pressure (BHP) which enhance the reservoir's injectivity. In addition to the thermal effect, the magnitude of pressure build-up is greatly affected by the fluid injection rate and the distance from the point of injection (Eshiet et al. 2013; Luo and Bryant 2014). Modeling results generally show that fracturing starts because the rock material can't handle the pressure from the fluid injection. But later on, fracturing caused by shear failure becomes more common. In addition, it is generally seen that some wells that were not connected to the injection point had their own fracturing events. The topic is a relatively new area of study, particularly with regard to the thermal impacts. The current state of knowledge regarding the injection-induced fractures that propagate in GCS is not yet comprehensive. The literature contains a significant degree of controversy regarding the question of whether fracturing is beneficial or detrimental to CO₂ sequestration (e.g., Thijs and Kok 2023; Lu et al. 2024). However, the integrity of the wellbore is of paramount importance for any subsurface operation especially when considering the injection of cold, high-pressure CO₂ into a warm, depleted reservoir as discussed previously.

To investigate this issue, experiments were conducted in the cement sheath due to J–T cooling. Small- and large-scale effective permeability tests were performed on two different cement classes (Class G and HMR+), using pure cement samples, cement–caprock composites, and casing–cement composites with CO₂ exposure (Nassan et al. 2024a, 2024b, 2024c). HMR+ cement refers to high magnesium resistant cement usually used to cement salt formations and differs from Class G in the chemical composition and grain size (finer than Class G). Fig. 7 depicts the results of one of the experiments showing that the permeability (measured with pure CO₂) of the casing–cement bond depends strongly on the exposure time (2 days vs. 38 days) at constant temperature of 14 °C. The effective pressure is the difference between the confining pressure applied on the sample, which is wrapped in lead foil and a rubber sleeve (the orange element in the Fig.), and half of the injection gas pressure on the inlet of the sample (applied on the cross-section area of the cement sheath). CO₂ is reactive with cement Class G in the presence of water, therefore, pure cement samples were either extremely tight to CO₂ or impermeable. As for the casing–cement composites, they were showing permeability within an acceptable range for wellbore cement within the time frame of the experiments. According to the recommendations of the API, the permeability of conventional wellbore cement should not exceed 2×10^{-16} m² to ensure sufficient sealing capacity and maintain wellbore integrity. As for HMR+ cement, it showed better performance than Class G cement in both pure cement samples, cement–rock salt composite, and casing–cement composites. The reactivity of HMR+ with CO₂ was very low compared to Class G cement even when tested in autoclave and was in continuous contact with carbonated brine for 6 months (Nassan et al. 2025).

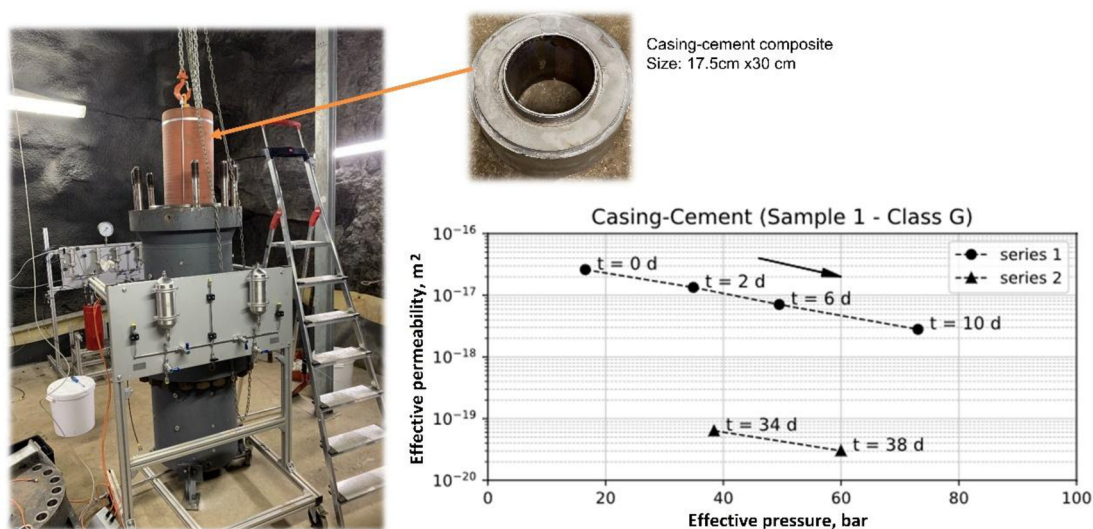


Figure 7—Large-scale permeability measurement setup (left hand side). Effective permeability of casing–cement composite vs. effective pressure at different time steps (all measurements were conducted at constant temperature of 14 °C).

To investigate the effect of temperature on casing–cement integrity, a series of experiments was conducted at an effective pressure of 40 bar, an initial gas pressure of approximately 20 bar, and inner casing temperatures ranging from 14 °C to –10 °C. Prior to this test series, the specimens had already undergone previous experiments at room temperature (14 °C), during which initial gas breakthrough had occurred. A total of three individual measurements were performed on several samples as follows:

- Stepwise reduction of temperature from 14 °C to 2 °C, –4 °C, –9 °C, followed by reheating;
- Stepwise reduction of temperature from 14 °C to –2 °C, –6 °C, followed by reheating;
- Measurement at ambient (room) temperature of 14 °C.

In comparison to the reference measurement at constant temperature, significantly smaller pressure changes—and thus reduced gas flow and permeability—are observed. One explanation for this behavior is that flow paths accessible to CO₂ at ambient temperatures are blocked due to freezing of pore water. However, this observation contrasts with the temperature measured at the outer radius of the specimen, which remains well above the freezing point of water. This suggests that the active flow paths are located in the immediate vicinity of the casing, at the interface between of material bond between steel and cement in this region.

The experiment was repeated on one specimen to verify reproducibility. Three cycles of freezing and thawing experiments were conducted on this specimen (Fig. 8). The figure shows that even at −9 °C, flow paths for CO₂ remained accessible. However, with an increasing number of cycles, permeability gradually decreased over time. In each case, a reference measurement was performed before and after the cooling cycle. The results also indicate that the contact time with CO₂ plays a role, which is attributed to the continuous chemical reaction between the cement and the CO₂. In summary, no detrimental effects of freezing on the structural integrity of the composite specimens have been observed thus far.

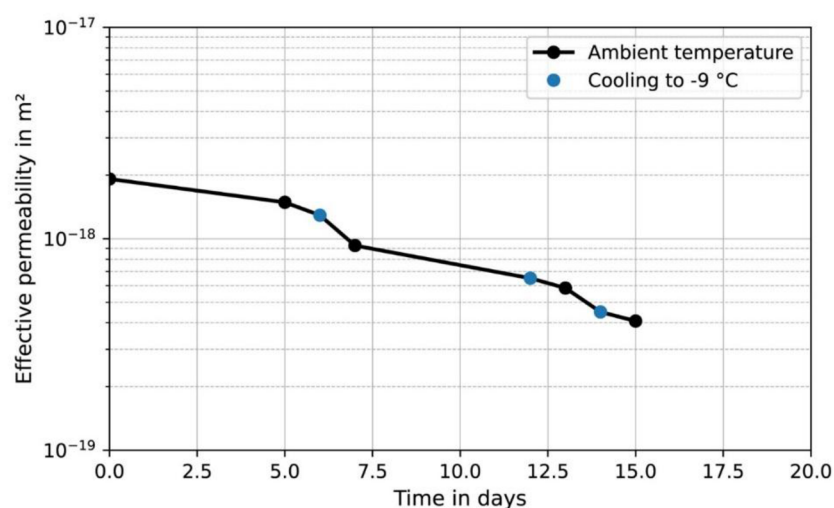


Figure 8—Effect of temperature cycling on the permeability of the second casing-cement composite specimen (after Nassan et al. 2024c).

Wellbore integrity under pressure cycling. Several composite casing–cement (Class G) samples were tested using different pressure cycling patterns. The pattern relevant to DGRs involves gradually increasing the pressure and then decreasing it. This pattern demonstrated that the composites become tighter—as indicated by a decrease in permeability—when the effective pressure is reduced. However, with repeated cycling, permeability increases and does not return to its original state (Nassan et al. 2024a).

Overall, the main findings of wellbore integrity under P & T cycling experiments can be summarized as follows:

- Experiments on pure cement samples (10 × 10 cm) of Classes G and HMR+ showed that both cement types were sufficiently tight for CO₂ storage. However, degradation of Class G cement was observed when CO₂ coexisted with brine in contact with the cement. This issue was not observed in HMR+ samples;
- Composite casing–cement integrity is highly dependent on the quality of the composite preparation; hence, on the quality of the cement job in the injection well;
- Subzero temperatures at −5 °C or −10 °C led to complete blockage of CO₂ flow at the casing–cement interface. Thawing restored flow without damaging the interface or causing micro-fractures;

- For casing-cement composites, the measured permeability was consistently less than $1 \times 10^{-17} \text{ m}^2$;
- Shale-cement (Class G) and anhydrite-cement (HMR+) composite samples showed extremely low permeability, suggesting that the cement-caprock interface is tighter than the casing-cement interface;
- The permeability of Class G cement was found to depend on exposure time to CO_2 ; where prolonged exposure time led to decrease in permeability.

Discussion and Mitigation

The Table 1 presents the results of our investigations focused on the challenges associated with using DGRs for CO_2 sequestration. The table also presents possible methods to be applied for their mitigation. The following discussion points are available for consideration:

Table 1—Challenges of CO_2 storage in DGR with impacts and mitigation methods.

Challenge	Impact, severity	Mitigation
Joule-Thomson	Additional cooling of the near wellbore, impact to all NWZ challenges, high severity	Assess and control well and near wellbore thermodynamics
Hydrate formation	Can plug the pores decreasing the injectivity, potential synergy with salt precipitation, medium severity	Avoid HSZ envelop, use effective inhibitors
Chemical reactions	Precipitation and/or dissolution of minerals, change in ϕ/k , longer term challenges, low severity	Assess the thermodynamic and chemical equilibrium with kinetics
Evaporation, salt precipitation	Can plug the pores decreasing the injectivity, increase in injectivity if evaporation standalone, low severity	Brine salinity and saturation are the main factor, assess and avoid cyclic operations
Thermal fracturing	Change in injectivity but also in well integrity, high severity depending on the magnitude of cooling	Assess and control well and near wellbore thermodynamics
Cyclic integrity	Reduction in mechanical strength of the casing/cement bond, low severity	Pre-investigation of stability and integrity under cyclic conditions
Hydraulic fracturing	Change in injectivity and integrity, high severity depending on the magnitude of cooling	Assess and control well and near wellbore thermodynamics

The formation of the J-T effect is observed to be contingent upon the coexistence of marginal thermodynamic and petrophysical conditions, namely the entrance of high pressure CO_2 into a reservoir characterized by low pressure and temperature, as well as low permeability. In combination with the cold CO_2 injection this can cause real harm especially in the NWZ. CO_2 hydrate formation and its negative impact on permeability and therefore on injectivity is strongly dependent on this rare combination. The most effective approach to mitigate the occurrence of hydrate formation is to prevent the HSZ delineated for the specific formation salinity. It is evident that as the salinity level increases, the HSZ decreases in size. In the event that hydrate formation appears to be inevitable under the prevailing conditions, the utilization of effective thermodynamic inhibitors becomes imperative.

The hypothesis that salt precipitation would effectively reduce injectivity by blocking the pores in NWZ appears to be implausible. This is mainly due to the low brine saturation in DGRs especially in the NWZ. As the evaporation capacity of CO_2 decreases with decrease in its temperature, the injection of cold CO_2 is another inhibiting factor. The possible stronger salt precipitation in the NWZ of the confining (under and/or overburden) formations due to their possible higher water saturations can be concluded as a positive aspect in terms of integrity and containment of CO_2 . However, in instances where the injection conditions do not align with the aforementioned positive conditions concerning salt precipitation—such as the implementation of warm CO_2 injection, elevated salinity levels, higher brine presence or influx, and/or low porosity and permeability—it is recommended to conduct a preliminary numerical assessment.

Following measures can be considered as mitigation measures against the above mentioned challenges of conducting GCS in depleted reservoirs:

Reliable numerical modeling of flow conditions in the wellbore and NWZ is essential for predicting thermodynamic conditions that could jeopardize injectivity due to the J-T effect. Estimating the J-T effect timely correct is the basis of all mitigation methods against the hydrate formation. If the thermodynamic conditions do imply the formation of hydrates under given petrophysics, proper mitigation chemical can be applied. An experimental study of the thermodynamic gas hydrate inhibition during high pressure CO₂ injection in depleted hydrocarbon reservoirs is provided by [Tamáskovics et al. \(2025\)](#).

There are no well-established mitigation methods that can mitigate the salt precipitation in NWZ of DGRs in the use of GCS. However, physical and numerical studies show that the risk of salt precipitation in DGRs is minimal due to low initial brine saturation. Confirming the brine saturation before beginning GCS operations with the help of field history and test results can ensure a precipitation-free operation. On the other hand, a potential brine supply through water influx from the gas-water interface can greatly increase salt precipitation. Therefore, designing the perforations in new drilled wells is of the utmost importance.

R&D Needs

Taking a holistic view of the challenges posed by using DGRs for GCS suggests that using DGRs still poses technical challenges that have not been well solved for general applications. It seems that the design and engineering of the CO₂ flow in wellbore and NWZ is of critical importance. The estimation of the thermodynamics at the bottomhole and NWZ under various operational scenarios including transient and cyclic modes is of paramount importance for the assessment of the technical challenges in the use of DGRs for GCS. Following R&D needs can be further elucidated:

- Modelling of the wellbore flow considering phase transitions and transient effects for better predicting the J-T effect in the NWZ.
- Physical and numerical modelling of the effect of hydrate formation on CO₂ flow in porous media.
- The effect of hydrate and salt precipitation on well integrity and well injectivity.
- The synergy between the hydrate formation and salt precipitation.
- Real-time monitoring of cement and casing/cement integrity during pressure and temperature cycling experiments using non-destructive methods, e.g., micro-CT or measurements of electrical and mechanical properties. Evaluation of the risk of cement cracking through long-term experiments addressing key well integrity issues.

Conclusions

Based on the above holistic review, the following results and recommendations for the use of DGR for GCS can be concluded:

- Notwithstanding the advantages inherent in this domain, significant challenges must be clearly delineated and overcome.
- The estimation of the thermodynamics at the bottomhole and at the NWZ is imperative for the assessment and management of the associated challenges.
- It is imperative to employ a capable and calibrated wellbore model coupled with the reservoir model to ensure the reliable estimation of NWZ thermodynamics, including the J-T effect.
- A preliminary check for possible hydrate formation is recommended considering all available data (gas composition, wellbore and near wellbore thermodynamics, formation brine salinity, petrophysics).

- In the event that thermodynamic conditions align with the HSZ, the hydrates can form from both liquid and gaseous CO₂. The most significant inhibitory effect is derived from the dissolved salts in water phase. In the context of a highly probable scenario, the availability of potential inhibitors should be ensured, and their capabilities should be thoroughly evaluated in a laboratory setting prior to implementation.
- Salt precipitation seems to be unlikely provided that no brine is supplied from the confining formations; however a preliminary numerical check is recommended. Salt precipitation in the NWZ of the confining formations containing higher brine saturations are likely and can be a positive aspect in terms of well integrity.
- Thermal and hydraulic fracturing should be addressed based on the assessments of well-reservoir coupled modelling. The effect of temperature on fracture initiation and development are the key issues, however the issue is case specific and needs to be investigated with geomechanical tests.
- Well integrity is a potential issue in DGRs; cement-CO₂-brine-rock interactions are case specific.
- Before starting CO₂ injection, it is recommended to ensure cased and cemented wells by wellbore logging. Furthermore, when a new cement is recommended, a lab measurement of its permeability to CO₂ and its integrity with casing and caprock must be ensured under the expected downhole operation conditions.
- Material compatibility with CO₂ is crucial for reliable and safe experiments.

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List of abbreviations

API	American Petroleum Institute
BHP	Bottom hole pressure
CH ₄	Methane
CIC	Capacity, injectivity and containment
CO ₂	Carbon dioxide
DGR	Depleted gas reservoir
DHR	Depleted hydrocarbon reservoir
DOR	Depleted oil reservoir
EOR	Enhanced oil recovery
GCS	Geologic carbon storage

Gt	Giga ton
HMR+	High magnesium resistant cement (fine grains)
HSZ	Hydrate stability zone
HYEX	Hydrate experimental setup
INJECTWELL	Experimental and Numerical Assessments of CO ₂ Injectivity and Flow Assurance During Storage in Depleted Hydrocarbon Reservoirs (Project)
J-T	Joule-Thomson
k	Permeability
Mt	Million ton
mD	Millidarcy
Micro CT	Micro computed tomography
N ₂	Nitrogen
NWZ	Near wellbore zone
PV	Pore volume
R&D	Research and development
RETURN	Reusing Depleted Oil and Gas Fields for CO ₂ Sequestration (Project)
SA	Saline aquifer

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